CCIT Report No. 352

Large area crack-free GaN on top of Si-on-insulator substrates

S. Zamir, B. Meyler and J. Salzman

ABSTRACT

GaN was grown by metal organic chemical vapor deposition (MOCVD) on plain Si on insulator (SOI) substrates, and on SOI which was pre-patterned for lateral confined epitaxy (LCE), with square mesas, separated by dip trenches. It was found that GaN films grown on plain SOI exhibit reduced crack density. Moreover, using LCE on SOI substrates dramatically increases the crack-free area of the layer. Areas of 100 μm × 100 μm were grown completely crack free, and stress was released by strong bending of the GaN/Si/SiO_2 composite islands.

Keywords: GaN on Si, Lateral Confined Epitaxy, Si-on-Insulator.